

Comparisons & Logistic Regression

Session 8

ISS Data Visualization with R
Faculty of International Social Sciences, Gakushuin University
Spring 2026

Plan for today

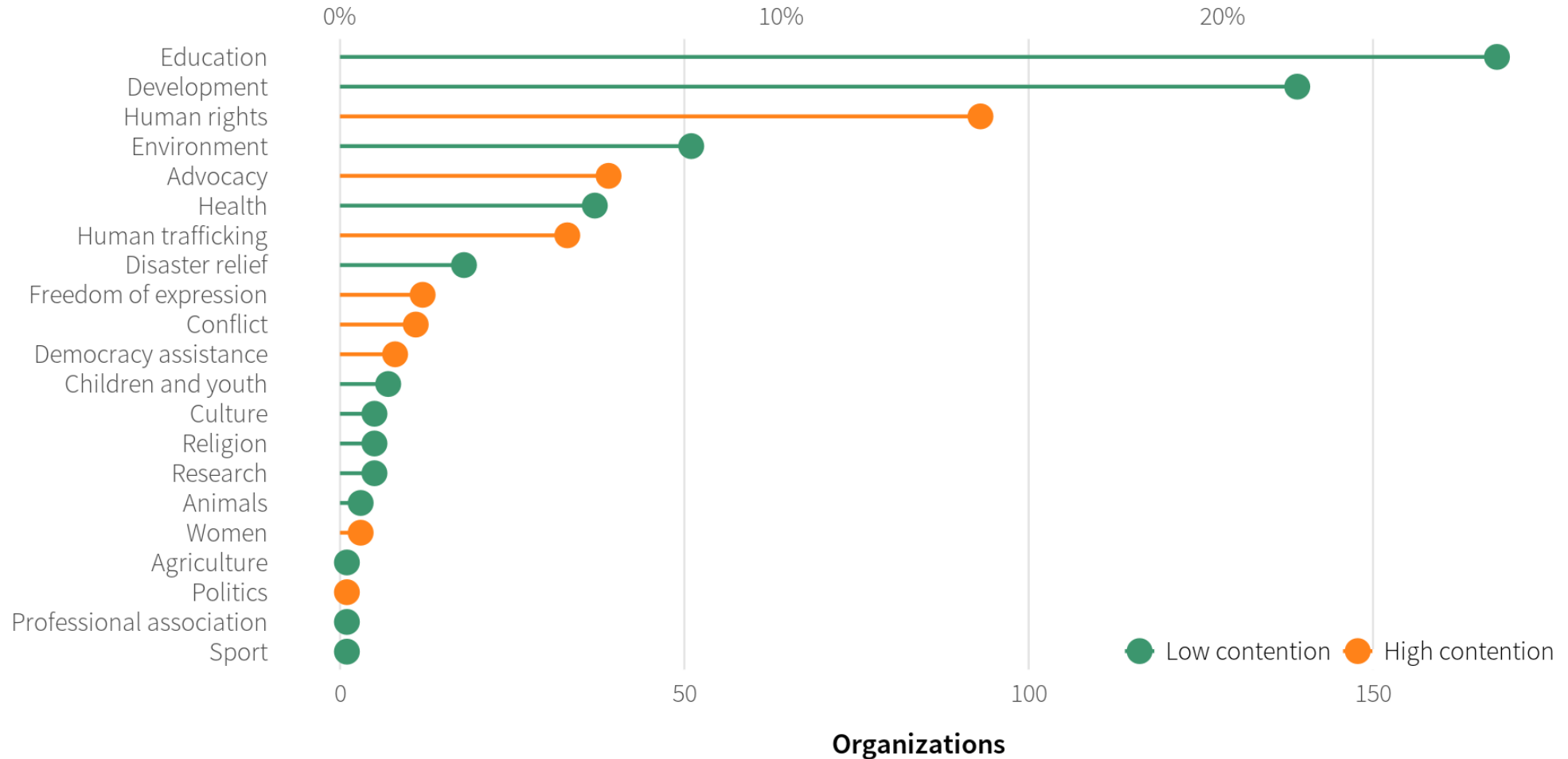
Visualizing comparisons

Small multiples as a bridge

Probability, odds & logistic regression

Visualizing comparisons

Lollipops and bars





Small multiples

How Trump compares with past presidents

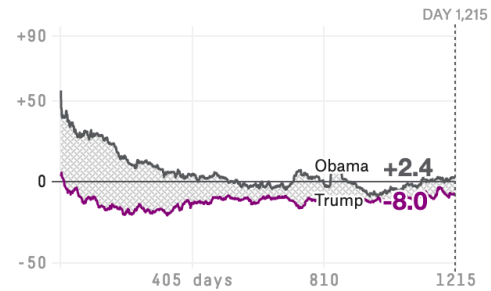
○ Approval rating ○ Disapproval rating ● Net approval

1,215 days

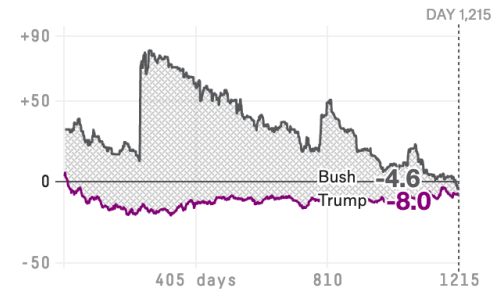
4 years

8 years

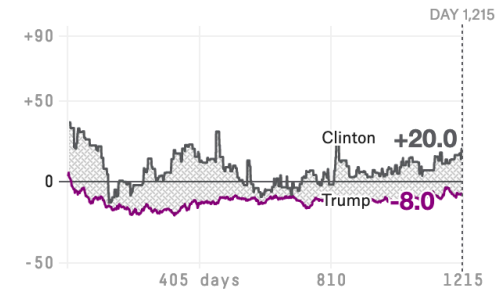
Barack Obama 2009-17



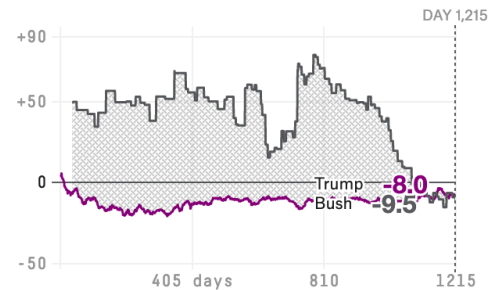
George W. Bush 2001-09



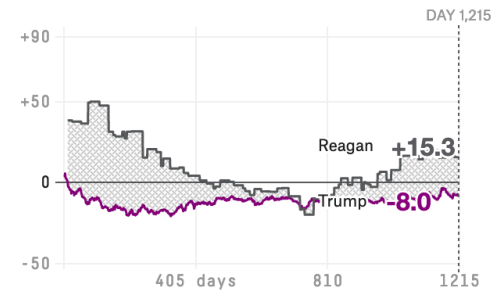
Bill Clinton 1993-2001



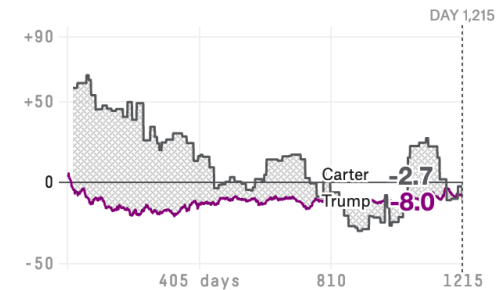
George H.W. Bush 1989-93



Ronald Reagan 1981-89

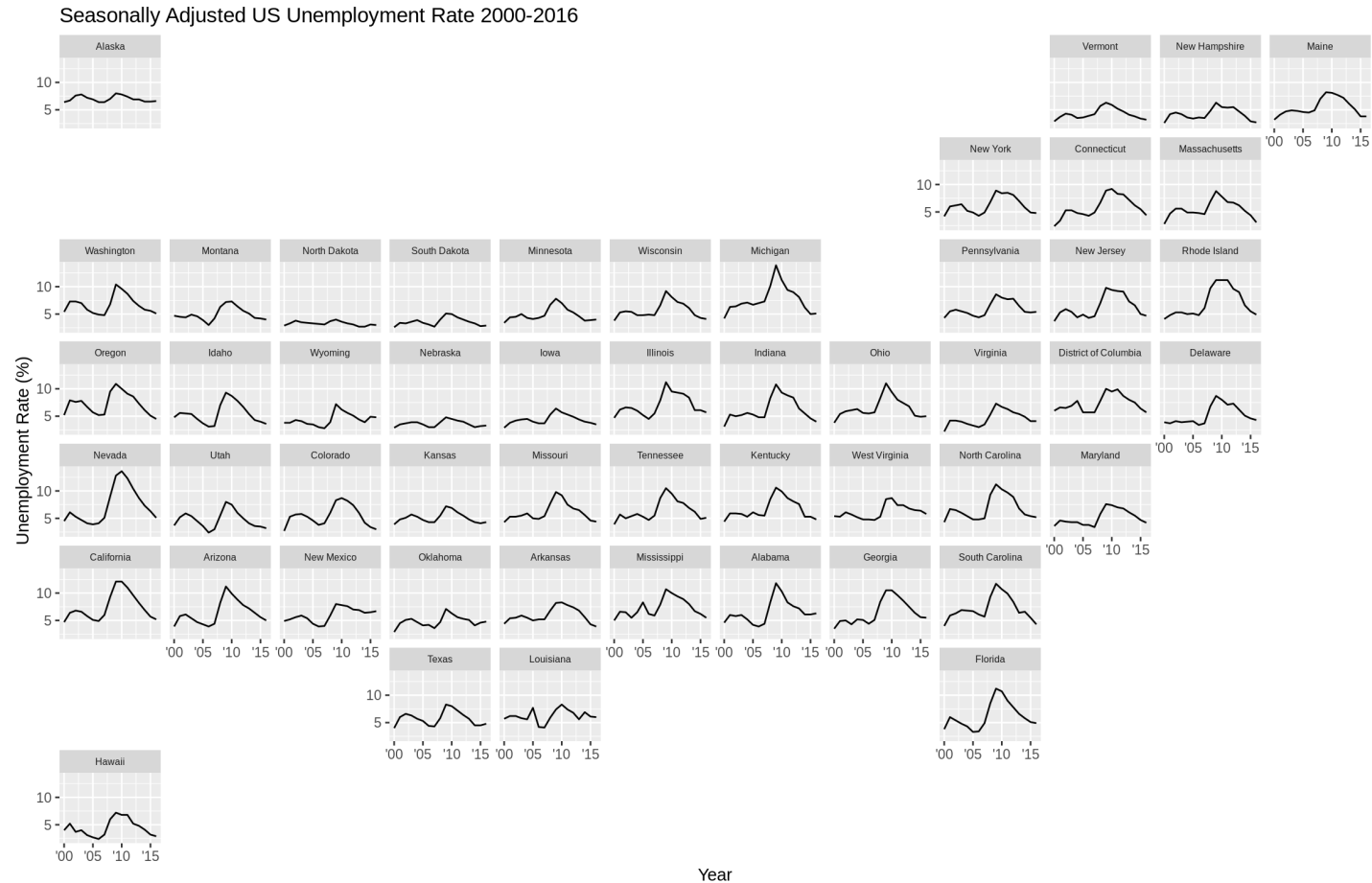


Jimmy Carter 1977-81



FiveThirtyEight, Trump approval ratings

Small multiples with larger shapes



`facet_geo()` in the **geofacet** package

Sparklines

Mauricio Pochettino has lead Spurs on their best run **8TH**  **2ND** in 24 years of the Premier League

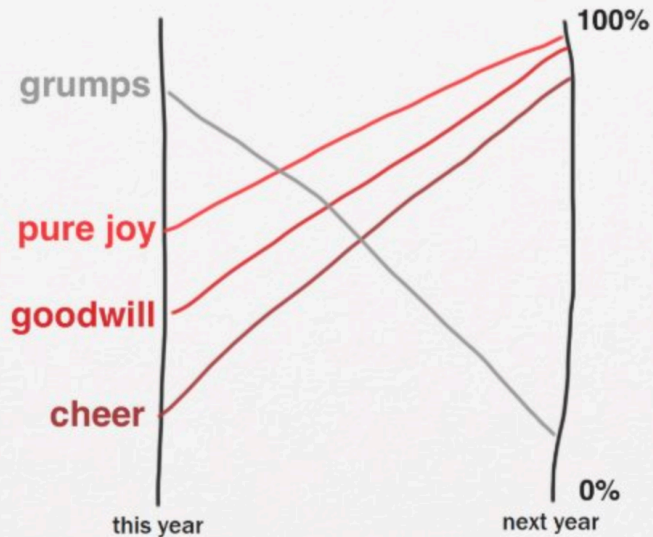
Alibaba stock is at 5 yr high **93.89**  **152.11** as of July 2017

The FTSE100 Brexit bounce **5562**  **7501** continues one year on from the vote last summer

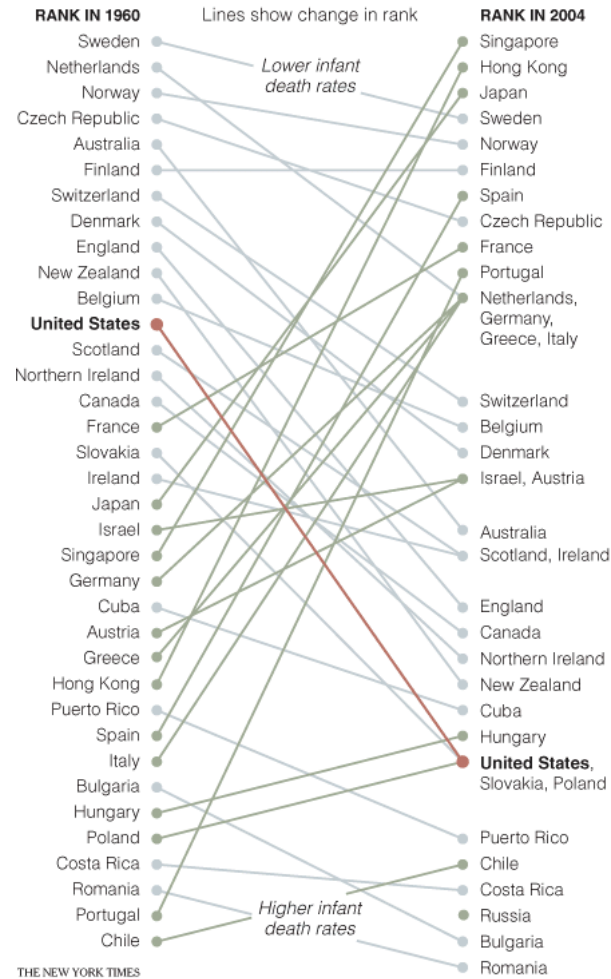


Slopegraphs

**May your New Year see
a significant increase in
all the good stuff.**



Slopegraphs

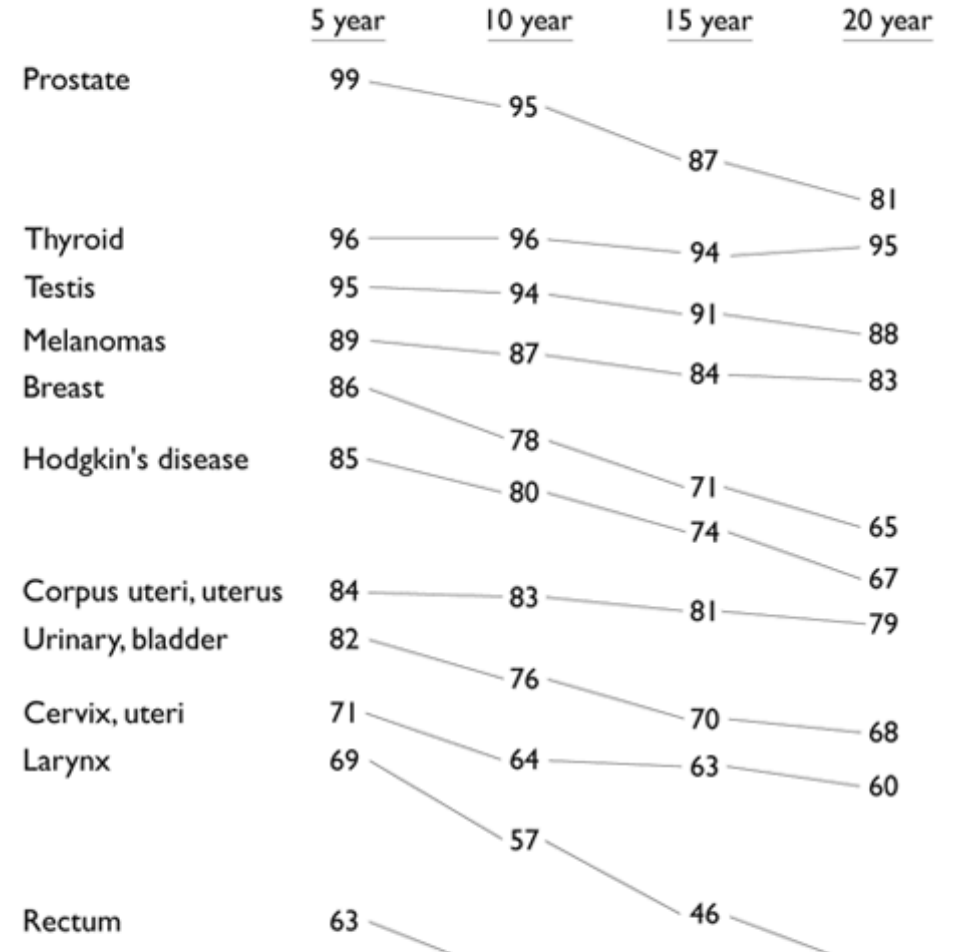


Slopegraphs

Estimates of relative survival rates, by cancer site

	% survival rates and their standard errors							
	5 year		10 year		15 year		20 year	
Prostate	98.8	0.4	95.2	0.9	87.1	1.7	81.1	3.0
Thyroid	96.0	0.8	95.8	1.2	94.0	1.6	95.4	2.1
Testis	94.7	1.1	94.0	1.3	91.1	1.8	88.2	2.3
Melanomas	89.0	0.8	86.7	1.1	83.5	1.5	82.8	1.9
Breast	86.4	0.4	78.3	0.6	71.3	0.7	65.0	1.0
Hodgkin's disease	85.1	1.7	79.8	2.0	73.8	2.4	67.1	2.8
Corpus uteri, uterus	84.3	1.0	83.2	1.3	80.8	1.7	79.2	2.0
Urinary, bladder	82.1	1.0	76.2	1.4	70.3	1.9	67.9	2.4
Cervix, uteri	70.5	1.6	64.1	1.8	62.8	2.1	60.0	2.4
Larynx	68.8	2.1	56.7	2.5	45.8	2.8	37.8	3.1
Rectum	62.6	1.2	55.2	1.4	51.8	1.8	49.2	2.3
Kidney, renal pelvis	61.8	1.3	54.4	1.6	49.8	2.0	47.3	2.6
Colon	61.7	0.8	55.4	1.0	53.9	1.2	52.3	1.6
Non-Hodgkin's	57.8	1.0	46.3	1.2	38.3	1.4	34.3	1.7
Oral cavity, pharynx	56.7	1.3	44.2	1.4	37.5	1.6	33.0	1.8
Ovary	55.0	1.3	49.3	1.6	49.9	1.9	49.6	2.4
Leukemia	42.5	1.2	32.4	1.3	29.7	1.5	26.2	1.7
Brain, nervous system	32.0	1.4	29.2	1.5	27.6	1.6	26.1	1.9

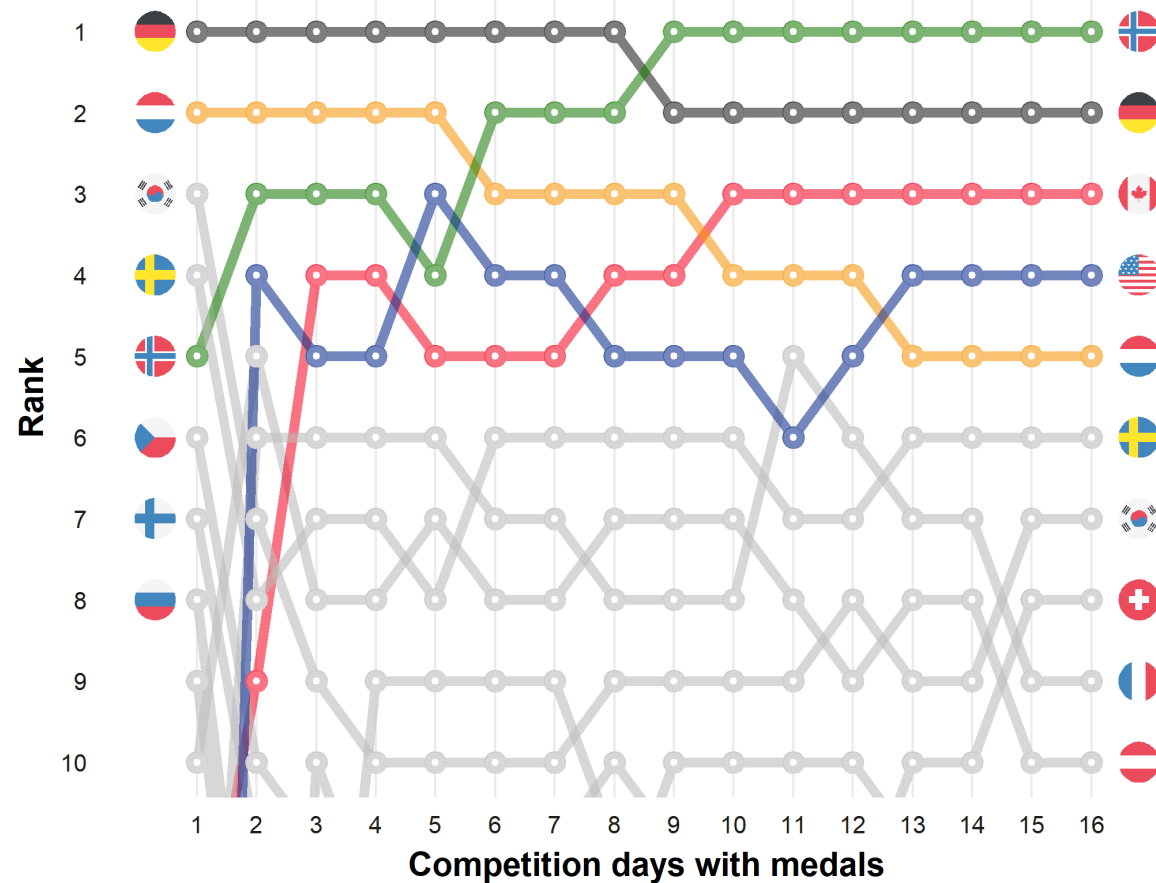
Estimates of % survival rates



Bump charts

PyeongChang 2018 Olympic Winter Games

Countries ranked by overall medals after each competition day



Small multiples

The core principle

One chart per group

Identical axes everywhere

Comparison is spatial

`facet_wrap()` in `ggplot2`

"Illustrations of postage-stamp size are indexed by category or a label, encoded with not more than two or three variables, and placed within the eyespan."

— Edward Tufte,
Envisioning Information

Why not one chart?

One chart — hard to read

Four groups, four colours,
overlapping lines,
busy legend — the comparison
is buried

Small multiples — easy to read

Same four groups, one panel
each —
differences jump out spatially

**The eye does the comparison work.
You don't have to decode a legend.**

Small multiples → logistic regression

We've seen small multiples for **comparing distributions**

They work just as well for comparing **model predictions**

A logistic regression with a **moderator** produces one predicted curve per group

`facet_wrap(~ group)` turns a moderation effect into a small multiples plot

Probability, odds & logistic regression

1. Start with probability

Probability (p)

How likely something is to happen

Ranges from **0** (never) to **1** (always)

Example:

75% chance of winning a game

But probabilities have a problem...

For regression, we need outcomes that can range from $-\infty$ to $+\infty$

Probabilities are **stuck between 0 and 1**

We need a way to **unlock** them

2. Odds

Odds compare the chance of something **happening** to it **not happening**:

$$\text{Odds} = \frac{p}{1 - p}$$

Example: if $p = 0.75$

$$\text{Odds} = \frac{0.75}{0.25} = 3$$

Key intuitions

Odds = 1 → equal chance (coin flip)

Odds > 1 → event more likely

Odds < 1 → event less likely

Odds are **always positive** — not quite what we need

3. Log odds

Taking the **logarithm** of odds gives something that can be negative, zero, or positive:

$$\text{Log Odds} = \ln(\text{Odds}) = \ln\left(\frac{p}{1-p}\right)$$

Examples

If Odds = 3 $\rightarrow \ln(3) \approx 1.10$

If Odds = 1 $\rightarrow \ln(1) = 0$

Why log odds?

Odds are always positive

Log odds can be **negative, zero, or positive**

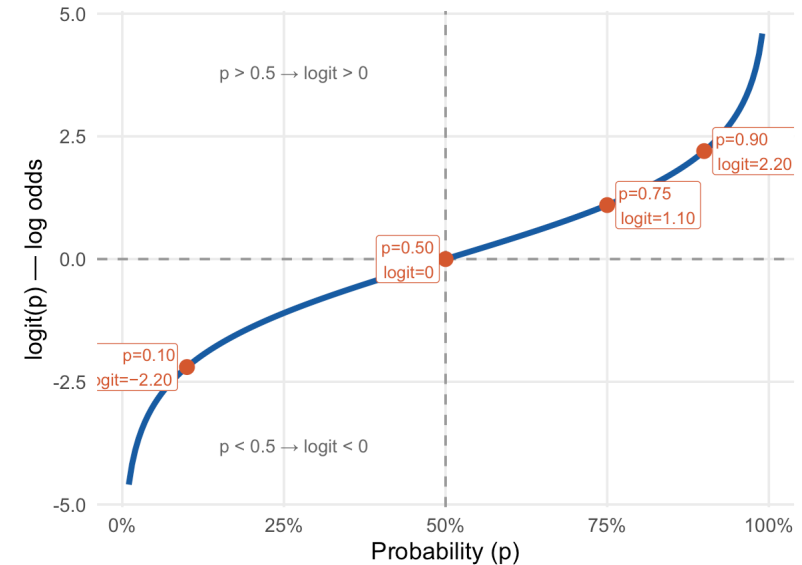
4. The logit function

The **logit** transforms probability into log odds:

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$$

Probabilities are **stuck**: 0 to 1

Logit **unlocks** them: $-\infty$ to $+\infty$



p	logit(p)

5. A worked example: free throws

A basketball player: **75% chance**
of a free throw $\rightarrow p = 0.75$

Odds: $\frac{0.75}{0.25} = 3$ (3× more
likely than miss)

Log odds: $\ln(3) \approx 1.10$

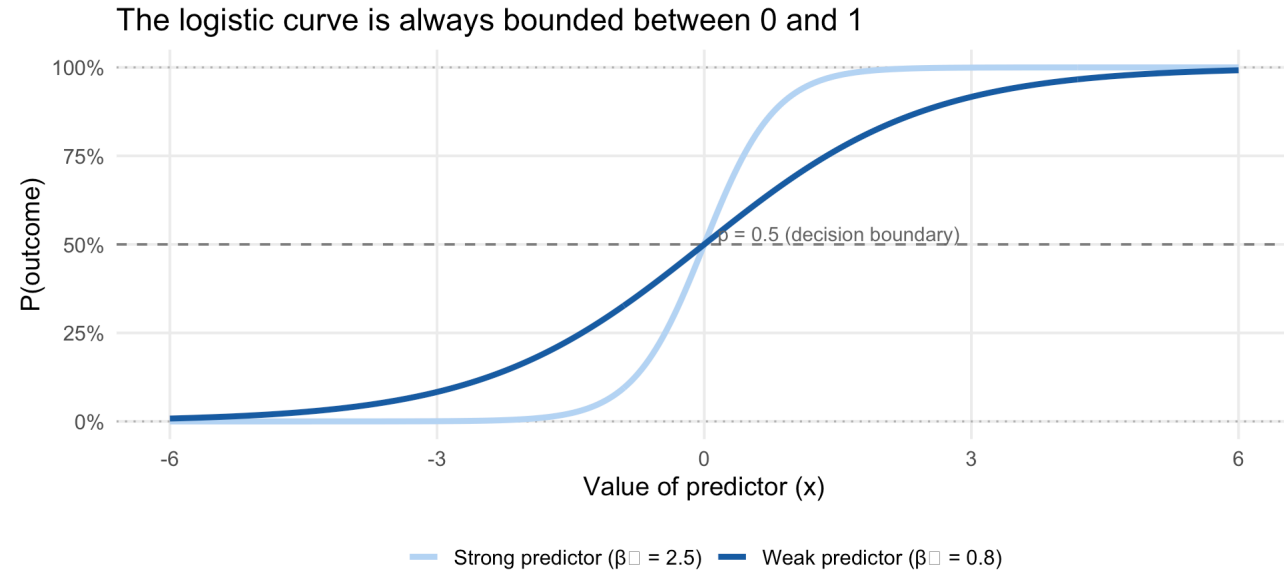
Three ways to say the same thing

Probability: 75% chance
(*intuitive, limited to 0–1*)

Odds: 3 to 1 in favour
(*success vs. failure*)

Log odds: 1.10
(*usable in regression*)

The logistic S-curve



6. The logistic model

For a binary outcome y and predictor x : $\text{logit}(p) = \beta_0 + \beta_1 x$

Because the logit is invertible: $p = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$

S-shaped logistic curve

Admissions example

Always bounded between 0 and 1

β_0, β_1 estimated via `glm()`

$$\text{logit}(p) = -12.04 + 4.08 \times \text{GPA}$$

Each 1-point GPA rise multiplies odds by $e^{4.08} \approx 59$

From theory to code

In R, logistic regression uses `glm()` with `family = "binomial"`

```
model <- glm(admitted ~ gpa,  
             data      = admissions,  
             family    = "binomial")
```

`glm()` = Generalised Linear Model

`family = "binomial"` = binary
outcome

`"logit"` link is the default

See the practical session for model
fitting, evaluation, confusion
matrices, and small multiples of
predictions